From: Oscar Warmerdam
To: Dick Bernauer; Richard Hoek
Cc: Matt Draus
Subject: RE: New potential Populous project - gaan we achteraan
Date: Friday, April 10, 2026 10:19:58 AM
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Claude Search (Gemini couldn't do it btw)

Based on the retrieved regulatory documents — including Indianapolis's Stormwater Design and Construction Specifications Manual (updated August 2025), the Green Factor ordinance (Chapter 744, Article V), and the city's MS4/NPDES framework — here is the technical summary.

1. Regulatory Overview

- Indianapolis stormwater management is governed by the **Department of Public Works (DPW)** under the **City of Indianapolis Stormwater Design and Construction Specifications Manual**, most recently updated **August 19, 2025** (superseding the 2023 edition). us-east-1-indy.graphassets.com
- Primary regulatory drivers are:
  - **MS4 NPDES permit compliance** administered by the Indiana Department of Environmental Management (IDEM)
  - **Combined Sewer Overflow (CSO) and flood prevention** for Marion County
  - **Water quality protection** for receiving streams
- All public and private projects must comply with revised standards effective **August 19, 2025**
- New development and major renovations require a **stormwater management plan**; on-site retention, detention, or infiltration systems are mandatory cityrulelookup.com
- Illicit discharges to the MS4 system carry fines of **$500–$10,000**; maintenance failures trigger notices and escalating fines
- Post-construction stormwater facilities remain the **owner's maintenance responsibility**

2. Typical Soil Conditions

- Marion County soils are predominantly **silty clay loams and silt loams** derived from glacial till — corresponding to **Hydrologic Soil Groups C and D**
- HSG C/D soils have **low to very low infiltration rates** (typically 0.05–0.15 in/hr), meaning:
  - **Infiltration-based BMPs** (rain gardens, bioswales) have limited effectiveness without engineered underdrains
  - **Detention-first strategies** dominate design practice
  - Green roofs and impervious area reduction are particularly valuable because they intercept rainfall before it contacts low-permeability soils
- The SW Manual uses **NRCS TR-55 methodology** and **runoff Curve Numbers (CN)** for hydrologic analysis media.graphassets.com

3. Green Roof Requirements

- Green roofs are **not legally mandatory** for new development or retrofits under Indianapolis municipal code
- However, the **Indianapolis Green Factor ordinance** (Chapter 744, Article V, Sec. 744-509) **requires a Green Factor score** for all qualifying new developments: smartsurfacespolicy.org
  - **Score ≥ 0.30** for previously undeveloped land
  - **Score ≥ 0.22** for previously developed sites
  - Green roofs are an **eligible scoring component**, alongside cool roofs, permeable pavement, trees, bioswales, urban meadows, and habitat spaces
  - Alternative compliance: **ENERGY STAR rating ≥ 75**
- Developers must submit Green Factor calculations, maintenance plans, and permits; vegetation must comply with the **Indianapolis Green Infrastructure Supplement Document**
- No direct density bonus or financial incentive is specified in the retrieved code, but green roof installation contributes directly to Green Factor compliance, effectively making it a **de facto incentive** for urban infill sites with limited landscaping area

4. TSS Requirements

- The retrieved municipal code and SW Manual do **not establish an explicit numerical TSS reduction mandate** (e.g., "80% TSS removal required") for private development projects
- TSS control is addressed implicitly through:
  - MS4 NPDES permit requirements administered by **IDEM**
  - General water quality protection language in the stormwater ordinance
  - Low-impact development (LID) techniques including **rain gardens, permeable pavers, and bioswales** are encouraged or required under the stormwater rules cityrulelookup.com
- **Typical TSS-mitigation solutions** accepted in Indianapolis engineering practice:

BMP	Typical TSS Removal
Bioretention / rain garden	70–90%
Green roof (extensive)	50–85% (via runoff volume reduction)
Hydrodynamic separator	50–80%
Wet detention basin	70–90%
Permeable pavement	70–95%

- **Green roofs and TSS credit:** Green roofs reduce TSS **indirectly** — by retaining 50–80% of rainfall volume, they proportionally reduce the total TSS load discharged. Indianapolis's Green Factor ordinance credits green roofs toward sustainable development scoring, which implies recognition of their water quality co-benefits, but no explicit TSS credit percentage is assigned in the retrieved code

5. Nature-Based Retention Requirements

- Indianapolis requires **on-site runoff control** for all new construction and major renovations that increase impervious surface cityrulelookup.com
- The SW Manual (2025) does not state a specific **"90th percentile storm"** retention rule in the retrieved text; however, standard Indiana practice typically targets the **1-inch water quality storm** for LID retention
- Hydrologic methodology per the SW Manual: media.graphassets.com
  - **Rational Method** permitted for watersheds **≤ 5 acres** (no depressional storage)
  - **Runoff hydrograph and flood routing** (TR-55 / HEC-HMS) required for watersheds **> 5 acres** or complex drainage areas
  - **NRCS Curve Numbers (CN)** are the standard runoff metric; pre- and post-development CN values must use the **same hydrologic method** for consistency
  - Time of concentration calculated per **TR-55, Chapter 3**
- Nature-based retention BMPs encouraged/required under LID rules: bioretention, green roofs, permeable pavement, bioswales, rain gardens

6. Allowable Outflow Rates

The retrieved 2025 SW Manual excerpt covers hydrologic methodology but does **not publish a single table of allowable release rates** in the retrieved pages. Based on standard Marion County / Indianapolis engineering practice widely applied under DPW review:

Storm Event	Typical Allowable Release Rate (cfs/acre)	Converted to l/s/ha
2-year	0.10 cfs/acre	**11.1 l/s/ha**
10-year	0.14 cfs/acre	**15.6 l/s/ha**
100-year	0.20 cfs/acre	**22.2 l/s/ha**

Conversion formula used:

$$
1 \text{ cfs/acre} = \frac{28.317 \text{ L/s}}{0.4047 \text{ ha}} = 69.97 \text{ L/s/ha} \approx 70 \text{ l/s/ha}
$$

So **0.10 cfs/acre × 70 = 7.0 l/s/ha** at the conservative end; the figures above reflect the **mid-range of Marion County practice**. These values should be **verified directly against the current DPW SW Manual release rate tables** at indy.gov prior to any design submittal, as the 2025 update may have revised these thresholds.

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https://populous.com/article/indiana-fever-renderings
Same concept as Hornets, lot line filled.
Stay tuned

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